

9/29 K3 Reading group

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From Degenerations to Periods: Moduli of K3 surfaces

§1 Motivation

Guiding Question

what does the moduli space of K3 surfaces look like?

Let X be a complex K3 surface.

We first study this question locally,
by deforming the complex structure on X .

§2 A Quick Tour of Deformation Theory

A deformation is a smooth proper holomorphic family

$$\pi: \mathcal{X} \rightarrow (S, 0) \quad \mathcal{X}_0 \simeq X$$

The Kodaira-Spencer map

$$\rho: T_0 S \rightarrow H^1(X, T_X)$$

associates to $\cdot \in T_0 S$: an infinitesimal direction
in the parameter space

• $H^1(X, T_X)$: the corresponding infinitesimal
change of complex structure.

Thus $H^1(X, T_X)$ is the intrinsic space of first-order
deformations

while $H^2(X, T_X)$ is an obstruct space.

For K3 surface, choose

$$0 \neq \sigma \in H^0(X, \Omega_X^2)$$

since σ is nowhere vanishing, contraction gives

$$T_X \xrightarrow{\sim} \Omega_X^1$$

$$v \mapsto \iota_v \sigma$$

Hence

$$H^1(X, T_X) \cong H^1(X, \Omega_X^1), \quad h^1(X, T_X) = h^{1,1}(X) = 20$$

and

$$H^2(X, T_X) \cong H^2(X, \Omega_X^1) = 0$$

Therefore the Kuranishi deformation germ $\text{Def}(X)$
is smooth of dimension 20.

We choose a sufficiently small simply connected representative

$$\pi_{\text{univ}}: \mathcal{X}_{\text{univ}} \longrightarrow S, \quad S \cong \text{a small open set in } \mathbb{C}^{20}$$

For the universal deformation,

$$T_S \xrightarrow{\sim} H^1(X, T_X)$$

So locally the moduli problem looks very simple:

$$\boxed{\dim_{\mathbb{C}} \text{Def}(X) = 20.}$$

§3. The Period Domain of K3 Surfaces

Why periods? : under deformation, we may think of the underlying differential manifold as fixed while its complex structure varies. Consequently, the topological cohomology H^2 remains fixed, but its Hodge decomposition

$$H^2(X_s, \mathbb{C}) = H^{2,0}(X_s) \oplus H^{1,1}(X_s) \oplus H^{0,2}(X_s)$$

moves with s .

For a K3 surface,

$$h^{2,0} = 1$$

Moreover $H^{0,2} = \overline{H^{2,0}}$, and $H^{1,1}$ is determined by orthogonality. Thus it is enough to record the single line

$$H^{2,0}(X_s) \subset H^2(X_s, \mathbb{C})$$

This is the period

Since S is simply connected, $R^2\pi_*\mathbb{Z}$ is trivial.

After choosing a marking

$$\varphi: H^2(X, \mathbb{Z}) \xrightarrow{\sim} \Lambda$$

We can identify the cohomology of every nearby fiber with the K3 lattice

$$\Lambda = U^{\oplus 3} \oplus E_8(-1)^{\oplus 2}, \quad \text{sign}(\Lambda) = (3, 19).$$

Hence the period map is

$$P: S \longrightarrow D, \quad s \longmapsto [\varphi(H^{2,0}(X_s))]$$

where $D = \{[x] \in \mathbb{P}(\Lambda_{\mathbb{C}}) \mid (x, x) = 0, (x, \bar{x}) > 0\}$

The first condition cuts out a quadric in $\mathbb{P}^2$, while positivity is open, so

$$\boxed{\dim_{\mathbb{C}} D = 20.}$$

The Hodge line varies holomorphically in a family, hence P is holomorphic.

§4 Local Torelli

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we now have two smooth 20-dim spaces

$$P: S \rightarrow D$$

Local Torelli Theorem

$P: S \rightarrow D$ is a local isomorphism.

It suffices to show that

$$dP_0: T_0 S \rightarrow T_{[0]} D$$

is an isomorphism.

- On the source,

$$T_0 S \simeq H^1(X, T_X)$$

- For the target, differentiating

$$(\sigma, \sigma) = 0$$

gives

$$(\sigma, \alpha) = 0$$

Therefore

$$T_{[\sigma]} D \simeq \text{Hom}(\mathbb{C}\sigma, \sigma^\perp / \mathbb{C}\sigma).$$

By Hodge decomposition,

$$\sigma^\perp = H^{2,0}(X) \oplus H^{1,1}(X),$$

$$\text{so } \sigma^\perp / \mathbb{C}\sigma \simeq H^{1,1}(X) \simeq H^1(X, \Omega_X')$$

Under these identifications, dP_0 is precisely contraction with σ

$$\begin{array}{ccc} T_0 S & \xrightarrow{d_0 P} & T_{[\sigma]} D \\ \downarrow s & & \downarrow s \\ H^1(X, T_X) & \xrightarrow{-\sigma} & H^1(X, \Omega_X') \end{array}$$

But $T_x \xrightarrow[\sim]{\cong} \Omega_x'$ [5]

Hence dP_0 is an isomorphism, and the holomorphic inverse function theorem proves Local Torelli.

Thus

periods give local coordinates on the deformation space of a K3 surface.

§5 From Local To Global

To globalize the construction, periods of different K3 surfaces must be placed in the same lattice.

Marking

A marking is an isometry

$$\varphi: H^2(X, \mathbb{Z}) \xrightarrow{\sim} \Lambda$$

A marked K3 surface is a pair (X, φ) , with

$(X, \varphi) \sim (X', \varphi')$ if there exists $g: X \xrightarrow{\sim} X'$ such that $\varphi \circ g^* = \varphi'$

Let $N := \{(X, \varphi)\} / \sim$

Around (X, φ) , take a small simply connected universal deformation S . The marking extends over the family, and Local Torelli allows these deformation spaces to serve as compatible local charts.

Thus N is a 20-dimensional, possibly non-Hausdorff, complex manifold.

Moreover, a marked K3 surface has [6]
no nontrivial automorphism preserving the marking,
since $\text{Aut}(X) \hookrightarrow O(H^2(X, \mathbb{Z}))$.

In fact, N is a fine moduli space of marked K3 surfaces.

The local period map glue to the global period map

$$P: N \rightarrow D, (X, \varphi) \mapsto [\varphi(H^{2,0}(X))]$$

By Local Torelli, P is a local isomorphism.

A deep global theorem gives

$P: N \rightarrow D$ is surjective

Thus every admissible Hodge structure of K3 type on Λ is realized by a K3 surface.

So far

$\dim N = \dim D = 20$, P is local isomorphism
and surjective.

one might therefore expect $N \cong D$

This is false.

§6 Atiyah's example and non-Hausdorffness

Consider Atiyah's family of quartic over a disk Δ .

$$x^2(x^2-2) + y^2(y^2-2) + z^2(z^2-2) = 2t^2$$

For $t \neq 0$, X_t is a smooth K3 surface.

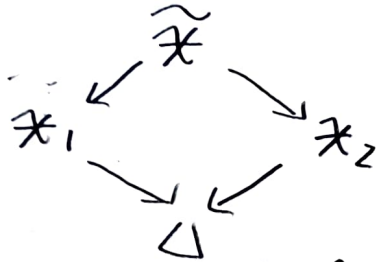
The central fiber X_0 has one ordinary double point,
and the total space is also singular there.

Its tangent cone is
 $x^2 + y^2 + z^2 + t^2 = 0$.

Blowing up the singularity gives an exceptional divisor

$$E = \{x^2 + y^2 + z^2 + t^2 = 0\} \subset \mathbb{P}^3 \simeq \mathbb{P}^1 \times \mathbb{P}^1$$

The two rulings of E give two different contractions



producing two small families

$$\pi_i : X_i \rightarrow \Delta$$

They agree over the punctured disk:

$$X_1|_{\Delta^*} \simeq X_2|_{\Delta^*}$$

but

$$X_1 \not\cong_{\Delta} X_2$$

This is the classical Atiyah flop.

After choosing compatible markings, we obtain

$$f_1, f_2 : \Delta \rightarrow N$$

such that

$$f_1(t) = f_2(t) \quad (t \neq 0)$$

$$\text{but } f_1(0) \neq f_2(0)$$

Hence these two points cannot be separated by disjoint neighborhoods.

N is non-Hausdorff

Their periods nevertheless coincide

$$P(f_1(o)) = P(f_2(o))$$

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So $P: N \rightarrow D$ is not injective.

Importantly, the underlying central K3 surfaces are isomorphic. What differs is the marked object.

§7 Global Torelli

Atiyah's example raises an obvious question:

If the global period map is not injective, does the Hodge structure nevertheless determine the underlying K3 surface?

→ Yes.

Weak Global Torelli:

$$X \simeq X' \iff H^2(X, \mathbb{Z}) \simeq H^2(X', \mathbb{Z}) \text{ as Hodge lattices}$$

Thus the integral Hodge structure with its intersection form determines the underlying K3 surface. Equivalently, after forgetting the marking

$$O(\Lambda) \backslash N \rightarrow O(\Lambda) \backslash D$$

is a bijection

This does not contradict Atiyah:

$P: N \rightarrow D$ remembers marked objects, whereas Global Torelli classifies the underlying K3 surfaces.

There is a stronger question. Given a particular [9]
Hodge isometry $\bar{\Phi}: H^2(X', \mathbb{Z}) \xrightarrow{\sim} H^2(X, \mathbb{Z})$
is $\bar{\Phi}$ itself induced by an isomorphism?

Strong Global Torelli

If $\bar{\Phi}(K_{X'}) = K_X$, then there exists a unique isomorphism $f: X \xrightarrow{\sim} X'$ such that $f^* = \bar{\Phi}$.

Thus

- Hodge structure determines the underlying K3.
- Hodge structure + Kähler chamber determines the geometric Hodge isometry.
- The period detect which classes are of type (1,1) but does not by itself distinguish which chamber of the positive cone is the Kähler cone

§8 Polarized K3 Surfaces

Fix a primitive ample line bundle with $L^2 = 2d$
and a primitive vector $l \in \Lambda$, $l^2 = 2d$.

For a marked polarized K3 surface require

$$\varphi(c_1(L)) = l$$

Since $c_1(L) \in H^{1,1}(X)$, $(\sigma, l) = 0$

Hence the period lies in

$$D_d = D \cap \mathbb{P}(l^\perp), \dim_{\mathbb{C}} D_d = 19$$

Let N_d be the moduli of marked Polarized $K3$ surfaces. We have $\boxed{P_d: N_d \rightarrow D_d}$

Global Torelli implies that

$\boxed{P_d \text{ is injective.}}$

It is, however, not surjective.

Suppose $S^2 = -2$, $(S, l) = 0$

and the period lies on

$$H_S = D_d \cap P(S_C^\perp)$$

Then S becomes a $(1,1)$ -class.

By Riemann-Roch, one of $\pm S$ is effective.

But $(L, S) = 0$, contradicting the ampleness of L .

Therefore

$$\boxed{\text{Im}(P_d) = D_d \setminus \bigcup_{\substack{S^2 = -2 \\ (S, l) = 0}} H_S}$$

Thus the (-2) -hyperplanes are exactly the obstruction to realizing a period with the chosen class l as an ample polarization.

§ 8 Final Picture

The polarized and unpolarized period maps have strikingly complementary behavior

	injective	surjective
$P: N \rightarrow D$	No	Yes
$P_d: N_d \rightarrow D_d$	Yes	No

For the unpolarized problem:

- every admissible period is realized
- injectivity fails because of the non-Hausdorff marked moduli problem.

For the polarized problem

- Global Torelli gives injectivity.
- Surjectivity fails precisely along (-2) -walls because of ampleness.

Finally, although Global Torelli gives a bijection

$$O(\Lambda) \setminus N \xrightarrow{\sim} O(\Lambda) \setminus D$$

this should not be confused with a well-behaved moduli-space isomorphism.

Forgetting the marking solves the set-theoretic classification problem, but does not magically remove the global pathology of the quotient.